

IoT Architecture

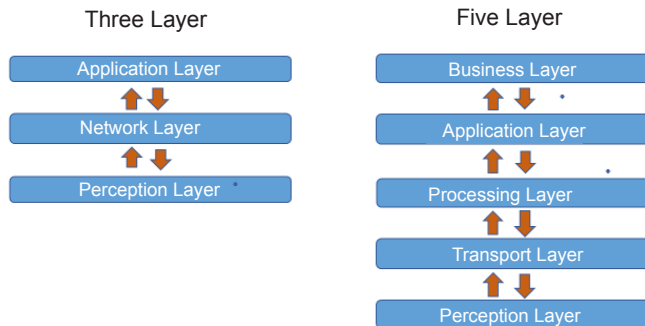


Figure 1: IoT architecture (Reference: <https://www.hindawi.com/journals/jece/2017/9324035/>)

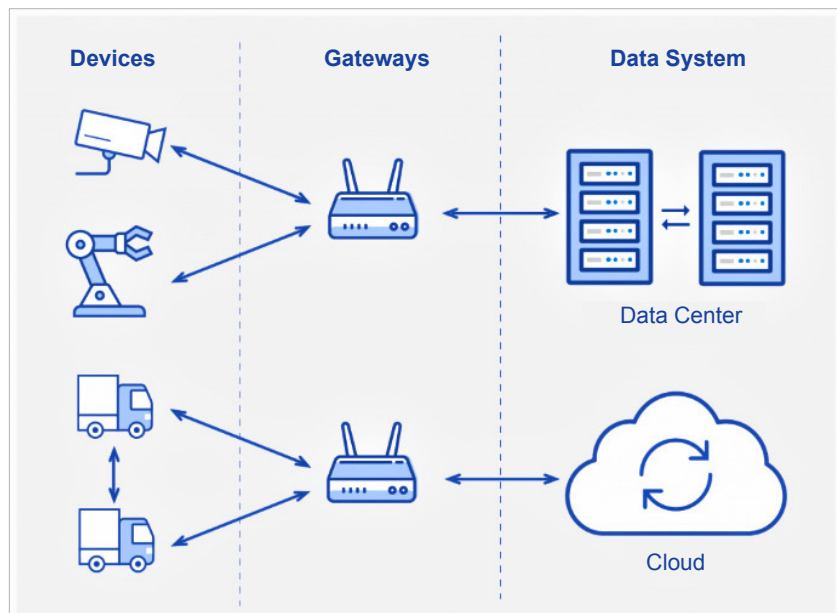


Figure 2: Types of IoT connections (Reference: <https://www.sam-solutions.com/blog/internet-of-things-iot-protocols-and-connectivity-options-an-overview/>)

various applications in which IoT can be deployed, like smart homes, cities, and health.

5-layer IoT architecture: The 5-layer architecture is an extension of the three-layer architecture with the addition of two more layers – the processing and business layers. The perception and application layers work in a similar manner as in the 3-layer architecture.

- The transport layer carries the sensor data from the perception to the processing layer and vice versa

using networks such as wireless, Bluetooth, 3G, RFID, and NFC.

- The processing layer or middleware layer stores, analyses, and processes large amounts of data that come from the transport layer, utilising many technologies such as databases, cloud computing, and Big Data processing modules.
- The business layer manages the whole IoT system, including applications, businesses, and user privacy.

Types of IoT connections

An IoT system has four types of transmission channels for data communication.

Device-to-device (D2D) communication allows the devices located in close proximity to communicate with each other using Bluetooth, ZigBee, or Z-Wave protocols. To establish a connection without a network is possible through a D2D connection.

Device-to-gateway communication connects devices using an intermediary platform. Gateways serve two main functions – first, to consolidate data from sensors and route it to the relevant data system, and second, to analyse data and if any problems are found, return it back to the device.

Gateway-to-data systems communication is the data transmission from a gateway to the appropriate data system.

Communication between data systems is within data centres or clouds. For this type of connection, protocols should be easy to deploy and integrate with existing apps. They should have high availability, capacity, and reliable disaster recovery.

There are two types of IoT protocols.

Network layer protocols: IoT network protocols connect medium to high power devices over the network. End-to-end data communication within the network is allowed using this protocol. HTTP, LoRaWAN, Bluetooth, Zigbee are a few popular IoT network protocols.

IoT data protocols: IoT data protocols connect low power IoT devices. Without any Internet connection, these protocols can provide end-to-end communication with the hardware. Connectivity in IoT data protocols can be done via a wired or cellular network. MQTT, CoAP, AMQP, XMPP, DDS are some popular IoT data protocols.